Higher National Diploma in Civil Engineering

2nd Year, Second Semester Examination 2019

CE2215-Structural Analysis

Questions

1.

I. Define **Castigliano's 1st and 2nd theorem** with the aid of a sketch. (04 Marks)

II. A simply supported beam XY of span **2L**, carrying a **UDL of w kN/m** as shown in Figure 01. Determine the **deflection at mid span** of the beam using **energy theorem**. Also clearly state any assumptions you made. (16 Marks)

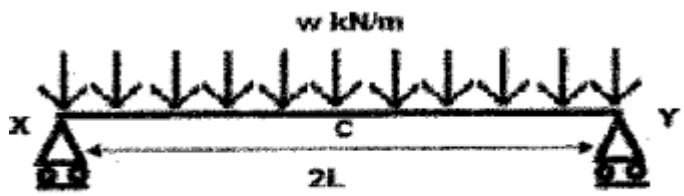

Figure 01

2.

I. Write the **three moment equation** of a two-span continuous beam and annotate the notations used in the equation. (05 Marks)

II. A continuous beam subjected to UDL and point load as shown in Figure 02. Determine the bending moment M_A, M_B ,M_C , and M_D using **three moment theorem** at supports A, B, C, and D respectively. Also plot the bending moment diagram. EI is constant throughout the beam. (15 Marks)

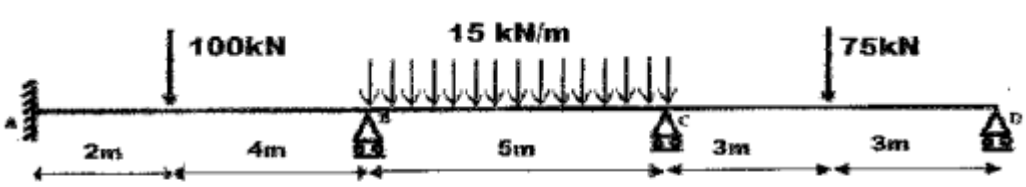

Figure 02

3.

I. State all the **assumptions** which are used in **slope deflection method**. (04 Marks)

II. Determine the **bending moment** of the continuous beam ABC as shown in Figure 03 using **slope deflection method** and plot the bending moment diagram. Also note that EI is constant throughout the beam. (16 Marks)

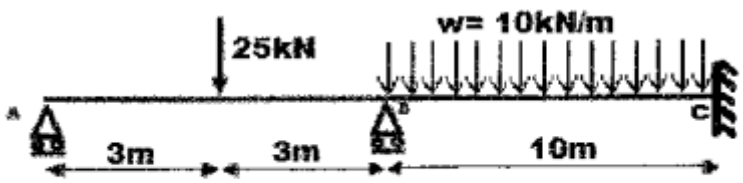

Figure 03

4.

A continuous beam ABC is carrying a **UDL of 20kN/m** and a **point load of 50kN** as shown in Figure 04. EI is constant.

I. Determine the **moment distribution factor**. (06 Marks)

II. Analyse the beam using **moment distribution method** and draw the **bending moment diagram**. (14 Marks)

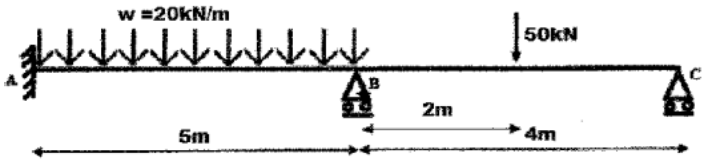

Figure 04

5.

I. Briefly explain why **truss members** are assumed to carry **only axial force** but no any moment transferred. (02 Marks)

II. Discuss the **two cases** in which member forces are **zero (zero force member)** in trusses with the aid of a neat sketch. (04 Marks)

III. Analyse the truss shown in Figure 05 using **method of section** and determine forces in the members **BC, BF, and FG**. (14 Marks)

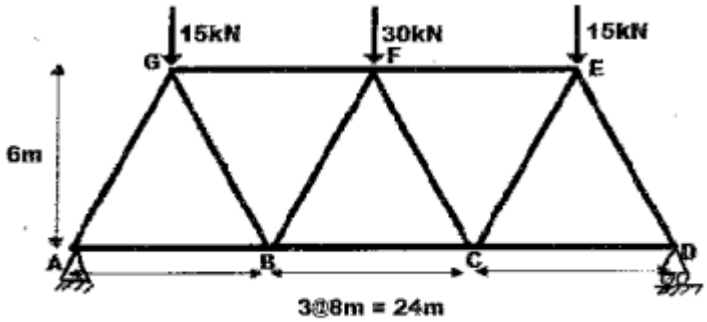

Figure 05

6.

I. Find the **horizontal reactions** of a **two-hinged parabolic arch** as shown in Figure 06. Neglect the shear and axial forces. (20 Marks)

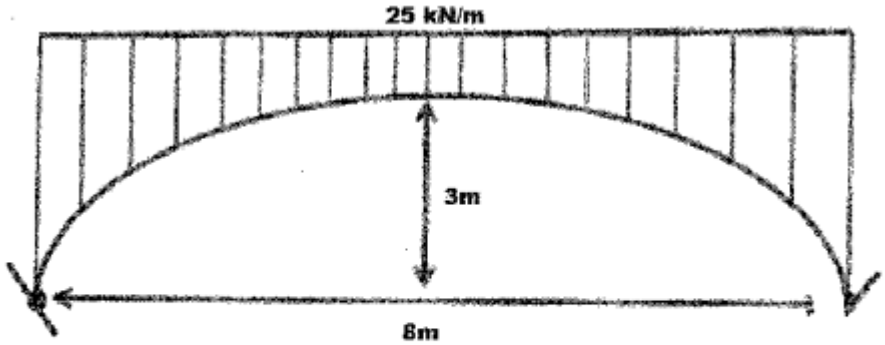

Figure 06

For parabolic arch $y_{(x)} = \frac{4hx(L-x)}{L^2}$ with usual notations